The Multichannel Reality of Discourse: Language-Paralanguage-Kinesics and the Totality of Communicative Systems

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ABSTRACT

Although verbal language is the most elaborate communication tool, treating it as an autonomous system has always hampered the study of many aspects of communication, for it occurs only accompanied by the two cosystems of paralanguage (voice modifications and independent word-like utterances coded in the vocal/narial-auditory channel) and kinesics (gestures, manners and postures), including the sign-laden nonactivities of silence and stillness. The three systems form the Basic Triple Structure of actual discourse (present in reading and represented or evoked by punctuation), made up, therefore, of verbal and nonverbal segmental and nonsegmental elements, and its multilevel transcription (indispensable for a realistic representation) shows beyond a doubt the fascinating cooccurrence of its components. But what truly constitutes the extreme intricacy of communication is the intimate interconditioning, not only of the speaker's own behavioral and even nonbehavioral activities, but of his and his cointeractant's verbal and nonverbal acts, being all qualified by their location in the communication stream, intensity and duration. A much needed workable classification of paralinguistic and kinesic phenomena is essential for a serious analysis of discourse. The inherent functions of paralanguage and kinesics are seen clearly when considering kinesic and kinesic-paralinguistic language markers, one of the three nonverbal categories, along with identifiers and externalizers, by which personal as well as cultural expressive styles are differentiated, and which simply cannot be ignored if we pretend to understand the reality of spoken language.

THE TRADITIONAL 'LINGUISTIC' APPROACH TO LANGUAGE

There is no need to emphasize that language, verbal language, is the *unique anthroposemiotic*, most elaborate and advantageous transactional tool and the backbone of human communication and culture. However, by blindly admitting this otherwise unquestionable truth, the very word language has been given a meaning which may seem either too broad or too limited in the light of a finer analysis of our communicative activities.

When we have given some attention to all that is generally not understood as language and yet communicates, the many forms of message-conveying systems which do not even have to cooccur with words, we may still speak of the superiority of verbal language, but could hardly separate it from all those activities. As we speak, accompanying, altering or alternating with the purely linguistic core of our deliverance, supporting or contradicting the essential messages conveyed by words and sentences with their stress and intonation contours, we develop other structured elements subtly interrelated which cannot be ignored. I am referring to the **paralinguistic** structure and to the **kinesic** one; and, from a cultural and interactive point of view, to all the other systems whose analysis should never be dissociated from that of the main triple body of activities, language-paralanguage-kinesics. One has traditionally isolated a verbal sentence claiming it represents 'a complete unit of meaning', distinguishing patterns of stress, pitch, and pausal phenomena, but nothing else. But a combination of phonemes and morphemes acoustically perceived would appear quite lifeless if we only attached to it the intonational features of stress, pitch, and juncture, for any stretch of speech under normal circumstances carries some extralinguistic elements (various voice modifications and independent sounds) which constitute paralanguage; and, if visually perceived, that sentence is accompanied by certain kinesic elements (gestures, manners, postures, and postural shifts), perhaps hardly noticeable, but intimately intertwined with both the linguistic and the paralinguistic costructures.

This is what I began to develop over ten years ago as the *Basic Triple Structure of Human Communicative Behavior* (Poyatos 1976, 1977a). For I soon realized, perhaps more so because at that time I was teaching only language and literature courses, that the study of language, even if we think only of the communicative activities developed in the vocal-auditory channel, should not be limited any more to the traditional 'linguistic' approach that concentrates on an arbitrarily pre-established set of segmental and suprasegmental elements but ignores whatever seems to lie beyond 'our' accepted phonological system or, in a very western attitude, shunts it as 'nonspeech', even though it is part of the phonological systems of lesser known languages. There are a great number of sounds perfectly normal from the physiological point of view, which as Pike (1943) complained are traditionally considered either as 'marginal sounds', because they do not seem to occur as phonemic norms in a given set of languages (whispered sounds, clicks, pharyngeal sounds, glottalized stops, ingressive sounds), or as 'nonspeech sounds', equally normal from the anthropophonetic point of view – but further away from traditional 'speech norms' (coughs, narial sounds, glottalized sighs, glottalized gliding vowels, etc.). Besides the fact that these sounds have been wrongly relegated to a status of abnormality and nonspeechness, and therefore excluded from phonemic charts, their semantic value within our vocal communicative repertoire is such that a study of the functions they play and

their 'lexical' value would certainly enrich, not just the anthropolinguistic field, but the other areas referred to in this article.

It also became apparent to me, in a logical process of dissatisfaction with the traditional limited perspectives in language and communication studies, that one could not carry out an in-depth semiotic-communicative analysis of language, vocal or nonvocal, from within a single discipline without unrealistically dissociating it, first of all, from a given cultural context, and secondly, from the mechanism of personal interaction. In culture we find a series of conditioning factors (geographical, ethnic, socioeconomic, psychological) and other extralinguistic elements that complement the semantic contents of verbal language (nonverbal vocal acts, gestures, spatial and temporal behaviors). In the social interactive mechanism typical of a culture we find a set of rules and patterns that definitely affect verbal language, both morphologically and syntactically, and gives it the characteristics of a true living communicative system.

THE EXPRESSIVE LIMITATIONS OF WORDS AND THE LEXICALITY OF THE BASIC TRIPLE STRUCTURE

It is clear that **words**, whether coined and utilized as arbitrary signs ('house') or echoic signs ('swish'), lack the capacity to carry the whole weight of a conversation, that is, all the messages being exchanged in the course of it. Our verbal lexicons are extremely poor in comparison with the capacity of the human mind for encoding and decoding an infinitely wider gamut of meanings to which at times we must refer as ineffable. If a natural conversation were to be conducted by means of 'stripped words' only (a concept hard to imagine), there would be, not just an intermittent series of 'semiotic (but signless) gaps', but some overriding vacuums as well, which are actually filled by nonverbal activities, either segmental (i.e., a click of the tongue, a sigh or a meaningful silence) or stretching over varying portions of our speech, from single phonemes to sentences to a complete conversation turn.

A single uttered word can act as an emotional display of anguish (' ¡God! ')[1], doubt ('Maybe'), fear (' ¡Oh! '), etc. – and even so, it needs certain prosodic features and specific paralinguistic modifiers (implied in the context or described in written language) – but rarely as an emotional-blend conveying all three feelings. But such an affect-blend could not be expressed either by means of a morphologico-syntactical arrangement of words – as one would not include in that sentence the lexemes 'anguish', 'doubt', and 'fear', to begin with – because it would entail an 'unnatural' complex periphrastic expression which just would not happen in such a state. What actually suffuses those words with life is a series of paralinguistic and kinesic elements subtly interrelated which support, emphasize, or contradict them.

As for **the written word** –which is mentally uttered as long as it is visually

perceived – it is farther removed from the reality of the verbal-nonverbal construct, for which reason the narrative writer refuses to just let his characters speak in a printed language without commenting on that language himself and describing its paralinguistic features. Written words are not just printed symbols on a piece of paper, and even the snow-white paper, the smelly rough one or the grayish or time-yellowed one do become semiotically interrelated with both the events in the story and its readers' attitudes and sensitivity. They are, in the first place, mentally (if not *sotto-voce*) uttered by the reader, who must ascribe to them a series of linguistic, paralinguistic, and kinesic elements – besides all the described, represented, evoked or between-lines situations which transcend the page and constitute an important part of the story. Besides, although it is true that writing and reading are predominantly intellectual activities, they are far from being limited to visual perception. Precisely because both writer and reader – in greater or lesser degree – attach to the verbal constructs they produce or interpret a series of nonverbal voice modification and, if dealing with literary characters, speech-accompanying body movements and positions (plus any other somatic activities and contextual elements of the explicitly or implicitly described objectual world), the coding of signs in written language become an intricate and often ambiguous task, much more so when the written text is, in addition, an interlinguistic-intercultural translation (Poyatos 1977b, 1981a, 1982b).

As for **lexicality**, the Basic Triple Structure is the only communicative complex in which an object of the tangible world or an abstraction can be indistinctively denoted by a word from our established lexicon, a paralinguistic construct, or kinesically. A bad odor, for instance, after being decoded in the brain, can elicit a verbal reaction ('¡That stinks! '), a paralinguistic one ('¡Eeugh! '), or a hand-to-nose gesture. Furthermore, considering the capacity of these three systems for mutual substitution within a preserved syntactical order (altered only from a 'linguistic' point of view), we can argue that the sentence '¡Oh, I feel – ! ' can be completed, rather than verbally ('¡Oh, I feel like kissing you! '), proxemically, changing toward one's interactant while displaying an appropriate facial expression; or that blushing can act as a cutaneous predicate of '¡Oh. I'm so – (embarrassed)! ', but this internal costructuration is never so deep as within the triple structure where verbal, paralinguistic, and kinesic expressions combine invariably in a live sentence.

We cannot neglect the fact that the elements of the Basic Triple Structure can be seen in a complex of bodily articulations as it would be possible to draw a human body and indicate the various articulatory parts where specific movements are formed, and the way in which they can be formed (i.e., point and manner of articulation). The phonemic-kinemic representation of the head, for instance, would be particularly interesting, for it would show: internally, the speech organs (teeth, lips, tongue [tip, blade, front, back], etc.), and externally, the facial moving parts

(eyebrows [high, middle, low, knit], eyelids [raised, lowered], pupil [dilated, contracted], mouth [degrees of closeness and openness, of distension and contraction]), etc. The display of paralinguistic qualifiers like labial control, maxillary control or nasal control (referred to the nose and not only to actual velar position) demonstrates the intimate interrelatedness of vocal and kinesic communicative behaviors. The whining gestures of wrinkled nose is congruent with and conditions the heavy nasalization typical of such a behavior. This leads us to ponder on the parallel developmental characteristics of the actual morphological, anatomical, and semantic interdependence of the speech and movement communicational streams.

When Kendon (1972:451), one of the most insightful students of nonverbal behavior today, expressed so rightly the obvious but neglected thought that "we must consider movement as well as speech if we are to understand what is entailed in what we somewhat loosely refer to as an act of speech" – and I would link this statement with those just made regarding the kinetic basis of communication – he confirmed the additional information provided by kinesics to what is being said (besides stress and grammatical markers). This shows that besides the very doubtful 'redundancy' of kinesics in most situations, it is also an **economy device**, as it 'says' something else in the same length of time. On the other hand, there is still another way in which kinesics can operate, namely, when the same person addresses two receivers simultaneously, one verbally and nonverbally and another one kinesically only, as when giving directions to a waiter without interrupting our conversation with others, or when someone who is engaged in a telephone conversation with one cointeractant initiates a simultaneous exchange with a second one by using a string of kinesic emblems to say (move) to him '/Com in/, /I'll be with you in a minute/, /Good to see you/, /sit down/, and /look at this in the meantime/', while listening to his speaker on the phone (cf. Johnson, Ekman, and Friesen 1975:342).

SEGMENTAL AND NONSEGMENTAL ELEMENTS WITHIN THE BASIC TRIPLE STRUCTURE

If we try to consider only a breath-semantic stretch of utterance – limited, that is, to the vocal-auditory channel involving language and paralanguage – as the most sensible division in speech (disregarding neither its physiological configuration nor its meaning or message) we soon come to the conclusion that it is possible to single out the vocal auditory band, so to speak, and that we need to include kinesic behaviors as well, from both a semantic and syntactic point of view. That breath-semantic group (similar, for instance, to the one shown in Figure 1, or much more elaborate) allows us to recognize the segmental and nonsegmental levels of sign production, that is, two types of activities:

(a) some which precede or follow each other as discrete portions of a noncontinuous whole: **words** (i.e., phonemes), **paralinguistic alternats** (i.e., vocal/narial nonverbal constructs, e.g., 'Uh', 'Tz-tz', a narial egression of contempt), **silences** or breaks in that audible chain of segmentable events, conversation **kinesic constructs** (i.e., gestures, manners, postures) which coincide or alternate with the audible ones (although some movements can be heard as well) and which are also discrete semantic segments, and **still positions** of one or all the parts of the body which interrupt the kinesic chain;

(b) others which clearly change throughout that communicative stretch of sounds and movements and silences and still positions, with not so clear boundaries and overriding those other elements, from syllables and/or simple kinemic constructs to much longer portions of speech or whole kinesyntactic complexes, varying slightly but with a cumulative impression never given by the clearly discrete parts, therefore not being segmentable: the elements we subsume under **intonation, primary qualities** (e.g., volume, tempo), **qualifiers** (e.g., maxillary control) and **differentiators** (e.g., overriding laughter), and the **parakinesic qualities** of intensity, range and velocity.

It is interesting to note that while two segmental vocal constructs cannot go together (e.g., word with silence, silence with an apico-alveolar click), vocal segmental ones can (e.g., a word with a gesture, silence with kinesics), just as vocal nonsegmental and kinesics (e.g., high pitch and gesture, low pitch and nonsegmental kinesics). This observation further proves the semantic and somatic cohesion of the Basic Triple Structure – without ignoring other cooccurring and conditioning (not just contextual) systems (e.g., dermal) – and totally justifies the study of punctuation, not just on the basis of language but as involving language, paralanguage, and kinesics.

This realistic view of a simple sentence poses two questions, besides showing how semanticity is both segmental and nonsegmental (which in turn reveals important problems with respect to written literature): the structural-semantic nature of intonation and that of silences and still positions.

Returning to the breath-semantic group, an expression such as 'Tz, ¡well, ¿what d'you think of that ? ! ', would show the following possible segmental and nonsegmental activities:

		3 *well*	*2* *What d'you think*	*4* *of that*
Language				
Paralanguage	segmental	Tz, pharyngeal ingression pauses after 'tz' and 'well'	junct. after 'think' drawled 'what'	drawled 'that'
	nonsegmental	drawled 'well' high rising volume high pitch	clipped 'd'you' same lower pitch	same higher pitch
Kinesics	segmental	raised brows smile hand stretched out	same same same	same same same
	nonsegmental	medium-tense movement wide-range brow raise med-to-wide smile med-speed movement	same same widening smile	same same wide-range smile with last high pitch and primary stress still facial expression

Figure 1. Segmental and Nonsegmental Levels of Sign Production

Only in looking at written language could we think exclusively of the vocal-auditory channel, since there is a cooccurrent kinesic (i.e., kinetic) activity formed by segmental linguistic-paralinguistic elements as well as segmental kinesic and parakinesic suprasegmental ones. Such an observation suggests, of course, the ontogenetic costructuration of the Basic Triple Structure. Furthermore, its high cognitive capacity, not inferior in a beckoning sign to its verbal counterpart, brings forth also the inevitable broadness of the concept of language, since it is clear that not only different messages, but a single one, can be – and most of the time is, if almost imperceptible – carried out in its totality only by the cooccurrence of verbal, paralinguistic, and kinesic behaviors.

In consequence, the segmental and nonsegmental elements of the Basic Triple Structure, that is, of direct face-to-face interaction, can be diagrammatically depicted as in Figure 2.

SEGMENTAL	Words	Paralinguistic alternants	Paralinguistic silences	Kinesic constructs	still positions
NONSEGMENTAL	Intonational features	Paral. Primary Qualities, Qualifiers, Differentiators		Parakin. Intensity, Range, Velocity	

Figure 2. Segmental and Nonsegmental Elements of the Basic Triple Structure

Again, the extreme communicative limitations of written language, most significant in narrative and dramaturgic literature, becomes obvious when we realize that only words, some ambiguously represented alternants ('Tsk-tsk', 'Ugh'), a few paralinguistic features ([!], [?], [( )], [CAPITALS], etc.), and some unmeasurable silences ([, ,], [. . .], etc.) can be present in a written text.

Two more points would be in order. One is that while segmental is mostly controlled by and conscious on the part of the emitter the same cannot be said of the nonsegmental behaviors, which seem to belong to two categories; namely, natural, that is, biological, either normal or pathological (e.g., a breath intake pause, a spastic movement), and controlled, whether personal and idiosyncratic or culture-specific, according to status, occupation (e.g., the auctioneer, the preacher), and situational context (e.g., reading to children). The other one is that, unlike sound and movement, which show correlates of silence and stillness, signs of a chemical, thermal, or dermal nature are opposed only by certain internal somatic nonactivities which cannot be apprehended by a receiver. Besides, the absence of those signs is rarely as relevant in social interaction as that of language, paralanguage, or kinesics.

CODING, INTERRELATIONSHIPS, AND QUALIFYING FEATURES OF VERBAL AND NONVERBAL BEHAVIORS

The first important basic aspect of personal interaction is that the receiver is usually more conscious of the speaker's nonverbal behaviors than the speaker himself, because nonverbal activities, from a paralinguistic click of the tongue, to blushing, or momentary interactive silence, may occur quite out of awareness. They may be, in other words, encoded and put in motion at a subconscious level, without the intellectual process required for the production of words, but decoded nevertheless by the listener. Furthermore, they can play important functions and have a definite effect on the decoding or interpreting process of even the verbal message when it is truly the central one, not secondary.

Given the inherent semantic limitations of words, the verbally encoded messages are fully decoded in natural conversation only when words are perceived and decoded along with their complementary nonverbal behaviors, a fact that demonstrates how unrealistic it can be to analyze meaning in terms solely of the morphological and syntactical constructs of verbal discourse. For encoding is not necessarily a conscious affair, and the signs – let us say words – emitted, as far as the speaker is concerned, reach the decoding end of the process 'colored' in a way that may not have been meant by their sender. Phonetic, semantic, lexical, morphological, and syntactical variations in the use of words and their prosodic features are tied to variations in the use of nonverbal modalities. Each activity can act either as **modifier** of one's own behavior or our cointeractant's, or simply as a **contextual** element. An apico-alveolar click 'Tz' + pharyngeal ingression can elicit my own 'Well . . . ', or slight frowning, or both, or it can make my interlocutor ask 'What it is?'; but it could be just a contextual behavior, as in 'Well, tz, I think I can do it.' Furthermore, that same click could modify the **meaning**, that is, the message (Tz + 'Yes', a rather negative tinge), the **form** (a hesitating sort of Tz + 'Yes') or the **behavior** itself (not eliciting a verbal expression but a shoulder shrug following 'Tz').

When affecting the meaning, a given behavior (a word, a click, a shoulder shrug or even a silence) can **support**, **emphasize**, or **contradict** the basic message. On the other hand – and this happens in conversation more often than we can imagine – my verbal or nonverbal behavior can modify my interactant's verbal or nonverbal behavior. 'Please don't quote me, but – ' can make him lean towards me, raise his eyebrows slightly, perhaps only blink slowly while staring at me; his blushing can (according to my interactive fluency and flexibility) give a new turn to my speech, or make me look away from him until his blush disappears, or smile.

Finally, both modifiers and contextual elements perform either a self-regulatory function or an interactional one, or both. One of my verbal or nonverbal acts can regulate (and not exactly modify), not only the production of my other verbal and

nonverbal activities – in terms of order of occurrence, amount displayed and beginning and end of the display (**self-regulatory** function) – but also my interlocutor's behavior – thus, regulating the flow of our transaction, by means of cues (words, pitch changes, gestures, silences) which indicate my intention to conclude the conversation, change the subject, contradict or agree with him, etc. (**interactional function**). Of these two functions, self-regulatory and interactional, the first one plays the most important role. By regulating one's multiple repertoire, it conditions in turn the cointeractant's behaviors, and the smooth or irregular rhythm of the conversation.

As for the qualifiers of interactive behaviors, costructuration, duration, and intensity, their analysis shows how unrealistic it can be to concentrate on a single message – conveying system, say, language, and attempt to discuss its communicative properties while neglecting its further relationships with the somatic and extrasomatic activities that may operate before, simultaneously to or after that system, closer or farther away from it; namely, somatic (physical/mental), environmental (subsuming living and objectual elements) and spiritual.

Costructuration refers not only to each of the speaker's behavioral or nonbehavioral activities with his other most immediate activities, that is, preceding, simultaneous and succeeding, but also to relationship with the verbal and nonverbal behaviors of his cointeractants[2]. Apart from the more obvious *a posteriori* effect of preceding activities, the simultaneous and succeeding ones are particularly interesting. As one of the aspects of the interactive functions, the speaker's silence, for instance, can be (and generally is, if imperceptibly at times) accompanied by movement (kinesics), just as his stillness can be (but not as often) accompanied by speech, thus conditioning each other and being affected, in turn, by their respective intensity, location, and duration.

As for the speaker's own succeeding behaviors, they, *a priori*, condition his other behaviors when he can predict them. For example, he prolongs his pause while intently looking at her, as he predicts the ensuing embrace. If he cannot predict that embrace, then the long, intense pause itself will elicit it (in fact, a blend of linguistic and/or paralinguistic, proxemic and kinesic complementary, and not redundant, messages). A similar process takes place for the costructuration of speaker and listener, except that the speaker cannot always accurately predict his cointeractant's behaviors.

Intensity depends here also on the duration of the behavior in question, for the longer a pause lasts, the tenser the whole situation may grow. While intensity refers mainly to the cooccurring activities that truly fill, for instance, the apparent gap of a silent or motionless segment of behavior, it can certainly be enhanced by the effect of the preceding behaviors, which determines, in turn, the characteristics of those occurring immediately after.

Duration, then, is ultimately related to both costructuration and intensity, as well as to what could be termed 'temporal signs' or chronemic attitudes.[3]

REDUNDANCY VS. COMPLEMENTARITY: PRIMARY AND SECONDARY SYSTEMS

One semiotic-communicative aspect of interactive human behaviors which I believe deserves the utmost attention, if some activities or systems are not to be neglected in the process of establishing their coding and interrelationships, entails a revision of the concept of redundancy (Poyatos 1981c).

That the concept of **redundancy** has been traditionally misused in everyday language as well as in serious research becomes obvious if we look at things with a semiotic eye. For what appears to be redundant is, most of the time, **complementary**, that is, supporting, emphasizing or contradicting the essential message. Blushing, for instance, is not necessarily redundant after or before words of the same message, for it may act as a dermal means of expressing the degree of emotion contained, or perhaps controlled, in those words ('I think I love you'), in their paralinguistic equivalent (a hesitant/affirmative closed-lips moan) or in their kinesic equivalent (a closed-eyes smile with averted gaze). It could also betray a deceiving verbal or nonverbal statement. Likewise, a beckoning gesture is complementary to 'Come here' when it specifically denotes how we really mean it, since the kinesic act can be performed in different ways. A silence, even after having verbally stated that we cannot say anything, adds information, as a complementary message, by its duration and by its costructuration with our kinesic behavior (perhaps our very stillness). One could argue, then, that redundancy is a fallacy most of the time, as the blushing or the gesture always assist the verbal or nonverbal message.

But redundancy must not be given a negative connotation of purposelessness either. Gesticulation, frequent pauses in conversation, the adoption of still positions, could certainly be redundant, yet express a personal style characterized precisely by redundancy. Thus, we might speak of redundancy on the communicative or message level, since nothing is being added, but hardly on the social level or the cultural one when a crosscultural difference is being observed. I heard some refer to the redundant quality of Italian or Spanish gesticulation as that kinesic quality which allowed them to identify those specific cultures!

This idea of redundancy *vs.* complementarity links with another needed differentiation in communication studies, that which exists between **primary systems** (or messages) and **secondary systems** (or messages), not to be confused with redundancy-complementarity. If we try to establish a **hierarchization** of systems in a given interactive situation – for each situation would entail a different organization of verbal and nonverbal behaviors – we soon discover that **the verbal code**, despite

the claimed superiority of 'language' over other systems, is not necessarily the main one. A rise in body temperature, blushing, or a silence, may carry the main message when language, paralanguage, kinesics or proxemics simply support it as complementary information or duplicate it as true redundancy. In this respect, what determines the hierarchization is the intensity of the various systems involved and their individual location in the communication stream.

Intensity is something we consciously or unconsciously measure by the degree of the qualifying features in each case, e.g., light or deep blush, mild or energetic negation gesture, mid-volume, normally articulated, level-pitched or shouted, tensely articulated, high-pitched verbal expression.

As for **location**, it refers literally to their physical position in the behavioral stream of the interactive situation. Tears can be the primary system (a chemical one) in relation to language if they give true shape to the main message. Silence and stillness may either precede or follow the most 'conspicuous' behavior, such as words or a gesture, but actually they may act as primary or secondary systems when judged by their location and by the characteristics of the related activities.

One could in human communication – not in zoosemiotics – refer to the criterion of cognitive control over the upcoming behaviors, which would modify all the factors just mentioned, and say that the controlled activities are primary ones while the uncontrolled ones are secondary, but this proves insufficient, simply because an uncontrolled display of tears, or our dermal perception of someone's coarse hand on our forearm, can certainly constitute primary systems carrying the most significant cluster of signs.

The redundancy-complementarity criterion, as well as the primary-secondary and controlled-uncontrolled ones, serve to enhance the internal costructuration of the Basic Triple Structure, language-paralanguage-kinesics. In addition, the triple transcription of language-paralanguage-kinesics discussed later can be an excellent tool to observe the exact hierarchization of systems in a single sentence, for instance, or in a breath-semantic stretch of speech. But, above all, it proves language as the purely anthroposemiotic system of communication.

PARALANGUAGE: NONVERBAL COMMUNICATION ON THE VOCAL/NARIAL-AUDITORY CHANNEL

If we try to compare verbal language and paralanguage, we see that verbal language, in the sense of the spoken string of words and sentences of human speech, shows, morphologically, (a) a **segmental** level or 'layer' formed by vowel and consonants phonemes, or smallest distinctive units (with their variations), combined to form morphemes (words, suffixes), or smallest meaningful units, which are themselves combined to form syntagms and syntactic constructurations; to that almost

lifeless body we must attach (b) a **suprasegmental** layer formed by what is commonly referred to as intonation, consisting of about four degrees of relative loudness, or stress, four different pitches, and three terminal junctures (rising, falling, level). These intonation patterns have no referential meaning in themselves, unless they qualify the lexical construct, although a paralinguistic, inarticulated, closed-lip expiration of air can convey different meanings, simply by varying its most important component, viz., intonation. We can say, however, that in a real communication situation, a phrase with its intonation contour is 'colored' by certain paralinguistic and kinesic elements, and that it is only then that many semantic changes and many otherwise impossible nuances are expressed; in other words, when that phrase reaches the totality of the interrelation of the three communicative activities which are the basis of unobstructed human interaction.

Paralanguage, simultaneous or alternating with those essential segmental and suprasegmental elements, offers the observer a series of equally segmental elements, but many more nonsegmental vocal and narial effects and sounds that cover a very wide range of acoustic phenomena determined, first of all, by the anatomy and the physiology of the individual's vocal tract, and secondly, by the idiosyncratic use he makes of those possibilities.

The problem with paralanguage is that it does not always offer a 'unit' analogous to the phoneme, susceptible of being built up into larger structures. All we can do with paralinguistic nonsegmental voice modifiers is abstract a specific quality of the voice, say, syllabic duration, and establish a scalar series of degrees below and above a medium line, that is, overclipped – clipped – medium – drawled – overdrawled; or velic control, which makes voice very nasal – nasal – medium – oral – very oral. Other features, which override actual speech admit, (a) a scalar classification of, for instance, giggling (which, at any rate, would also depend upon other features such as the velic control just mentioned, or pitch, thus showing also as many as five degrees in each case), or (b) a sort of 'class' ranging, in the case of laughter, from closed-lip muffled snickering to uprorious guffawing. But there is a fourth group ('alternants') which is independent from 'words' and which shows actual phonemes (like the apico-alveolar click, although not from the point of view of the English language) and also certain inarticulated sounds that are difficult to measure according to a base line (unless we consider their modifying accompanying features) and whose variations may or may not produce a semantic change.

I only summarize here the classification of paralinguistic phenomena I have offered before (Poyatos 1976) gratefully inspired by but drastically enlarging upon some pioneering work (mainly Pittinger and Smith 1957, Trager 1958, McQuown 1957, Pittinger, Hockett, and Danehy 1960; Austin 1965), on which I attempted to elaborate according to phonetic (i.e., phonological), semantic and functional criteria (in agreement with Crystal's [1974] excellent state-of-the-art paper).[4] This classifi-

cation acknowledges four well differentiated categories.

A. **Primary qualities**, (also referred to by others as 'voice quality', 'voice set', 'speaker identity', etc.) fundamental constituents of human speech which differentiate, first of all, a person from the others "over which most individuals have little or no measure of control" (Crystal 1971:198). I have classified as primary qualities: timbre, resonance, volume, tempo, pitch register, pitch interval, pitch range, syllabic duration, intonation range, and rhythm, recognizing conditioning factors such as **biological**, that is, purely somatic (such as sex and age, conditioning, for instance, timbre), **physiological**, thus variable, whether due to temporary malfunctions or to traumatized states (nasal resonance due to cattharr, improper timing in aphasias), **cultural** (e.g., higher volume of Latins and Arabs, certain dialectal peculiarities), **social**, such as status (the slow tempo of superiority), occupation (the orality of a preacher), or certain functions, like baby talk, story-telling, etc. In other words, primary qualities can also be based on nonpermanent factors that produce a sort of secondary or temporary voice set in a person. I will comment only on **volume**, that is, loudness, due to the respiratory and articulatory muscular effort, referred both to utterance or speech and to single syllables (beyond stressed or unstressed ones, e.g., 'It was *aw*ful'). Like pitch or speed, it possesses syntactical and cognitive value – as in a speeded-up low-volume (and pitch) parenthetical observation – and its parameter can go from very high (fortisimo) to very low (pianisimo), either rising (crescendo) or falling (diminuendo). Volume has an important culture-identifying value, as Arabs and Latin, for instance, are louder in streets and public places than the average AngloSaxon.

B. **Modifiers** are a series of vocal effects produced by factors like the direction and characteristics of respiratory air, by the way it is controlled in the vocal bands and how they vibrate, by certain changes in the pharynx, by the soft palate position, by how articulation is produced in terms of muscular tension and position of the organs, and by the anatomical configuration and shaping of the lips and the lower mandible. Some, of course, may appear as permanent traits of a person's voice, in which case they qualify words, differentiators, and alternants as true primary qualities, while others may be due only to temporary states or contextual circumstances, such as the labializing effect of baby talk, the ingressive utterance of verbal expressions of terror or the traumatized nasality of certain malfunctions. Some are reported as being phonologically used in certain languages, and others only paraphonologically, and most of them modify from syllables to long portions of speech, while a few others modify the verbal utterance occasionally (e.g., laughter), or appear isolated from it, in which case they must be regarded as alternants, the next major category of paralinguistic phenomena.

B.1 It is within **qualifiers** (respiratory-, glottis-, laryngeal-, velar-, pharyngeal-, articulatory-, labial-, and maxillary-control, and articulatory tension) that we first

encounter the typical ambiguity, from a phonetic point of view, of most of the descriptive or definitory labels employed, the utter lack of accurate physiological descriptions to refer to those effects, and the complete absence of orthographic forms to indicate at least impressionistically some important attitudinal features (just as we utilize exclamation marks, or repetition of letters to symbolize drawling). Although they can also appear as primary qualities, they do not show clear parameters in which to identify different degrees of a continuum, as they usually involve more than one anatomical and/or physiological change in the laryngeal or pharyngeal areas, for instance. Ideally, each one ought to be analyzed in terms of: anatomical and/or physiological configuration (e.g., relaxed vocal band vibration, weak arytenoid closure); auditory effect (e.g., nasal twang); voice type it produces (e.g., nasopharyngeal, creaky, breathy); cooccurrent verbal and nonverbal behaviors, as some qualifiers may be part of a whole established construct (e.g, pursed lips + lowered brow with irritated '¡Oh, leave me alone! '); phonological use (e.g., glottal stop in some West African languages); paralinguistic use (e.g., glottal stop for scorn, '¡Terrible! '); abnormal occurrences (e.g., hoarse voice of trachyphonia); and notation for phonetic purposes and because the core of the message may sometimes be carried by a qualifier.

B.2 **Differentiators** (laughing, crying, coughing, sighing, sneezing, belching, yawning, hiccoughing, etc., when they cooccur with words) characterize psychological and physiological states, and are closely costructured with kinesic behavior while being modified by primary qualities and qualifiers, since even reflexes like sneezes, belches and hiccoughs condition specific accompanying kinesic hand behaviors, for instance, according to cultural norms. Among them, laughter and crying deserve much research actually as paralinguistic-kinesic constructs, as regards: the biological foundation (age, sex, some sexual deviances); the influence of the psychological configuration on the frequency of occurrence, duration, acoustic characteristics and eliciting factors, as well as temporary emotional states and their relations to cultural display rules; pathological varieties; positive and negative social functions (e.g., affiliation, tension relief, comicality, grief, love, deception); the phonetic and kinesic variants according to the socioeconomic and cultural characteristics of the person; laughter and crying in noninteractive situations; their simultaneous or alternating costructuration with language, kinesics, proxemics, chemical (e.g., tears), dermal (e.g., blushing) and thermal (e.g., body temperature raise) messages; and the study of definitory references to and descriptions of laughter and crying in the narrative literatures of the various cultures.

C. **Alternants** (Poyatos 1975) can be described, in an impressionistic way, as sighs, throat clearings, clicks, pharyngeal ingressions and egressions, egressive frictions, hissing sounds, narrial frictions, moaning sounds, closed- or open-lip sounds, meaningful silences, etc. Each language, each culture, or social community possesses

a great number of perfectly encoded alternants, as they constitute a true lexicon used constantly in personal interaction as systematically as dictionary items, which some are already. They deserve, therefore, a much higher status in linguistics as well as in any disciplines dealing with communication, and the following basic facts muct be considered: (a) that they differ radically from other paralinguistic phenomena because of their lexical value, that is, their segmentality; (b) that they cannot be regarded any more as 'nonspeech' or 'marginal' sounds just because many are not constructed with phonemes from the best known languages; (c) that they play important roles in the mechanism of interaction, with as high a frequency rate as verbal items and often with an even clearer semantic and regulatory function (e.g., an apico-alveolar click + pharyngeal ingression as a turn-claiming cue, or signifying 'Don't say that'); (d) that they seem to form, more than 'words', the greater part of the communicative repertoire each culture utilizes for the interaction of people with mainly domesticated animals (see, e.g., Bynon 1976); (e) that they should be given serious thought with respect to the phylogeny of the Basic Triple Structure; and (f) that we need to increase drastically the present limited system of phonetic symbols, labels (i.e., verbs and nouns), and written forms, as we have for a few ('H'm', 'Psst', 'Er', 'Uh-hu'). While professional writers make their description of alternants a characteristic of their insight and literary style (Poyatos 1977b, 1981a), they also, like the layman, try very hard to represent them.

KINESICS: NONVERBAL COMMUNICATION ON THE KINETIC-VISUAL/TACTILE CHANNEL

When applying to its analysis the methodology of linguistic structuralism, Birdwhistell (1952, 1970) found in conversational kinesics a smallest discrete element, the **kineme** (analogous to the phoneme), made up of various **allokines** (similar to allophones); kinemes combine into morphological constructs called **kinemorphs** (analogous to morphs), forming **kinemorphemes** (comparable to morphemes) and **kinesyntactic constructions**. What is more, some suprasegmental elements, namely, kinemes of stress and juncture, have been reported by Birdwhistell (1970) as appearing in the linguistic-kinesic stream (while the relationship between linguistic pitch and body movement is under study). In addition, **parakinesic** degrees of **intensity**, **range**, and **velocity** (akin to stress and articulatory tension, syllabic duration and speech tempo) have also been identified, and they differentiate personal styles and in general all kinds of normal and abnormal visually-perceived behaviors as much as paralanguage differentiates audibly-perceived ones, therefore being as important to kinesics within the Basic Triple Structure as paralanguage is to verbal language. A foreign language, or a regional variety of the native one, can be spoken fluently, but usually with a certain accent which is revealed as inexact placement of

stresses, pitches and junctures and incorrect articulation of some vocalic and consonantal sounds (that is, inaccurate internal movements). In a similar manner, a foreign kinesic repertoire, or a regional variety, can be imitated, learned and performed with more or less fluency, but perhaps with an inevitable alien 'accent' in many instances, consisting in inexact placement of kinesic stress and junctures and, more noticeably, in the incorrect use of the parakinesic qualities (intensity, range and velocity) and the misplacement of certain kinemes within kinemorphemic constructs.

Kinesics as Total-body Articulation

It is an unquestionable principle that social interaction depends on sound and movement, and when either of those dimensions is lacking in man's emission or perception capabilities, normal life is severely hampered. That lexicality, based on verbal language, is increased by the conceptual possibilities of the joint language-paralanguage-kinesics structure, however. On the other hand, both sound and movement can not only be perceived directly, and the signs conveyed by them decoded audibly and visually, but also interpreted through other channels by our intellectualized synesthesial abilities that greatly enlarge our relationship with other human beings and the whole universe.

What no study of kinesic behaviors of any one part of the body must overlook is that the relationships between, for instance, the face and other parts of the body must be established at all times. One can set out to observe, analyze, and describe the different expressions of the face in a given situation, but the concomitant movements and positions of the legs, hands and arms, and the whole trunkjmust not be judged solely as independent 'contextual' elements, for it is the whole body that becomes articulated along with the face in that particular situation. It is possible, for instance, that we might record someone's brow raising + smile of surprise as someone else enters the scene, but it is hard to believe that a person under normal circumstances would not uncross his arms or legs if they were crossed, or lean forward at least slightly, in which case the articulation that constitutes the construct 'happy surprise' engages face, trunk, legs and/or arms, and hands. This would be the truly complete 'kinesic phrase' that will precede words of greeting or act by iteslf. In the same situation the lack of body movement, except for the face, may convey physical impairment or true absence of happy surprise, the passive attitude of the limbs and trunk betraying indifference at the very least. It is this total-body articulatory complex that illustrates the unparalleled possibilities and semantic nuances of anthropokinesics over zookinesics.

The fact that cultural advancement and technology have enriched the repertoires of gestures, manners, and postures through the evolution of the dwellings, of furniture, utensils, clothes, etc. (which also betray the progress of social and intel-

lectual life) can be attested today by systematically observing the kinesic repertoires of the urban and/or educated groups with those of the less educated peasants. The repertoire of the rural class is much more limited in gestures, manners, and postures, as well as in paralinguistic expressive means (as it lacks the versatility of subtly different types of laughter, of meaningful narial egressions, or closed-lip vocalic sounds) and vocabulary. In fact, this seems to emphasize the cohesiveness of the Basic Triple Structure from a functional, interactive and historical point of view, and the assumption that not just verbal language, but the three cosystems, have evolved along with the different cultures, since, the more primitive the culture, the more limited those repertoires seem to be.

On the other hand, what von Raffler-Engel (cf. 1980) is currently studying as 'developmental kinesics' (see also Peng and von Raffler-Engel 1978 and Hoffer and St Clair 1981) ought to be applied to the total linguistic-paralinguistic-kinesic complex and, very specifically, to the acquisition of paralanguage, particularly alternants.

Without going into the nature of consciousness in humans as compared to animals (cf. Thorpe 1972) we must consider how in humans, in spite of their also rapid fading, visually perceived movement can be said to travel through time. A derisive smile or an intent static look can linger and linger and truly regulate our subsequent behaviors and thoughts as the stored visual image is replayed over and over in our memory. We can also reconstruct the physical appearance of a person by gradually fitting together language, paralanguage, and kinesics until that triple structure brings the person to life in our minds, a process central to the existence of the fictional character (Poyatos 1977b, 1981a). And, with no mental effort whatsoever, one may easily relive the physical presence of someone else by carefully seeing and hearing with 'the eyes and ears of the memory', as has been described so many times in literature.

As for the morphological-perceptual characteristics of kinesics, it is very important to make a clear distinction, so often neglected, between: **gestures**, conscious or unconscious body movements made mainly with the head, the face alone or the limbs, learned or somatogenic, and serving as a primary communicative tool, dependent on or independent from verbal language, either simultaneous or alternating with it and modified by the conditioning background (e.g., smiles, eye movements, a gesture of beckoning and other emblems, a tic); **manners**, somatogenically modified but mainly learned and socially codified according to specific situations (e.g., the way one eats, greets others, coughs, crosses arms or legs to adopt those postures); and **postures**, more static (although the still posture proper is preceded by an important dynamic forming manner) and used less as a communicative tool, although they may reveal affective states and social status (e.g., sitting, standing).

Another morphological distinction that carries important sociopsychological, cultural, and historical implications is that between free and bound movements and still positions, which occur in both interactive and non-interactive situations. By **free** I understand a kinetic act or position performed by one or more parts of the body or limbs in space, that is, by themselves, without contacting other parts nor assisting themselves of any props provided by the objectual world. Free gestures (in interaction) are, therefore, the movements of the eyes, eyelids and the different parts of the face, head nods, shakes and tilts, and the gestures made with hands, arms and legs. Free manners are, in interaction, the way we bow in greetings, or stretch our hands; and in noninteraction, the way we walk (without trying to elicit someone else's behavior, as a female may typically do), yawn, or stretch. Free postures are stance, squatting, and any position in which only the feet are touching the sustaining surface. By **bound** I understand any movement or position in which the hands come in contact with other parts of the body or with each other, or in which any part of it comes in contact with other bodies or, mainly, with objects. Bound gestures, in interaction, are self-adaptors like touching one's temple to indicate 'He's crazy', and object-adaptors like pointing with a pencil. Bound manners are, in interaction, self-adaptors, like preening, scratching or rubbing one's eyes; object-adaptors, as when handling fork and knife, jewelry, pipes and glasses, or when knocking on a door; alter-adaptors, as in grooming, protecting, punishing; in noninteraction, they are self-adaptors, as in preening, object-adaptors, as in handling tools or playing an instrument, and body-adaptors, as in chewing gum, masticating food or drinking. Bound postures in interaction are, apart from standing, sitting and lying, the ones assumed mainly as alter-adaptors of aggressive or affiliative nature, of sexual behavior, etc.; and as object-adaptors, as a female's provocative way of leaning against a table toward her male interlocutor (perfectly costructured with her language and paralanguage, and even with chemical messages such as those emitted by perfume); and, in noninteraction, the object-adaptor positions of stance (against a wall), sitting and recumbence (on furniture, on natural objects, or as conditioned by clothes or by the characteristics of a vehicle), and those adopted in certain occupational activities in which desks, counters, mechanical artifacts and tools are engaged.[5]

THE TOPIC OF LANGUAGE MARKERS[6]

Language markers are conscious or unconscious behaviors which we all use in greater or lesser degree to punctuate and emphasize the acoustic and grammatical succession of words and phrases according to their location and relevance in the speech stream. They also coincide with written punctuation symbols, which are grammatical and attitudinal themselves. They have a semantic function only inasmuch as they identify grammatical functions, not the conceptual or affective

contents, which are identified by other nonverbal behaviors. While language markers have observed and defined as kinesic only, close analysis of their occurrence reveals that other bodily behaviors simultaneous with speech truly mark language, particularly paralanguage, thus completing the basic language-paralanguage-kinesics communicative structure. Furthermore, discourse is also marked, although in a more limited way, by proxemics, and even certain chemical and dermal reactions at times, while object-adaptor behaviors are, of course, part of kinesics.

Kinesic language markers enhance the syntactical functions of words, punctuate them much as we do in writing (thus giving punctuation visual and audible form), and coincide with paralinguistic features of identical function. Their function, then, is primarily grammatical (i.e., syntactical and morphological), although the 'full' semantic value of a word or combination of words is given, of course, by their paralinguistic and kinesic coactivities and any other cooccurrent behaviors – which only proves the coherence of the verbal and nonverbal channels. The following grammatical markers can be differentiated:

Pronominal markers are typically signalled by pointing with the hand(s), with a multiple head movement involving gaze (pointing even toward an absent human or nonhuman grammatical subject or object) and a tilt of the head, with a head tilt alone, or with a trunk-and-head orientation shift toward the referent. In many cultures chin-pointing toward another person for 'he', 'she', 'them', etc., is too familiar or low-class usage, never part of a more formal or refined interactive style. Head-pointing is accompanied by gaze-pointing when indicating our single cointeractant, but when two or more are present eye-contact may be (not necessarily but typically) maintained with the main addressee while the head points at a third party, (cf. Jespersen's [1933] "pronouns of pointing").

In any case, it must be emphasized that kinesic pronominal markers, however subtle (from gaze shift to a head-and-orientation pointer) are practically always present as part of the verbal-paralinguistic-kinesic structure. A typical instance of kinesic pronominal marker would be (John, Tom and Mary are conversing, and Tom addresses John): 'You should take her/slight head tilt toward Mary while looking at John/to the movies because I think she'll/head and gaze turned toward her/love that/very light head nod/movie'.

The preceding example serves to illustrate another function of the same markers, that of stress markers, since 'her' (Mary) and 'that' (movie) can carry also the two primary stresses which would be marked by the same object pronoun and demonstrative adjective.

The pronominal (often also adjectival) markers we use in conversation can be: personal (e.g., 'We can all/sweeping inclusive hand movement/ contribute'), **demonstrative** (e.g., 'These/inclusive glance and head sweep/are the best'), of **identity** ('Two years later I went there and met the same man/short up-and-down hand

movement, typically joining thumb and first finger in many cultures/'), of **totality** ('Every girl/double hand up-and-down movement/he met fell in love with him'), **possessive** (e.g., 'You can take our/sweeping pointing finger/bags now'), **indefinite** (e.g., 'Either/pointing at two referents in succession/'; 'Any/head or hand pointing sweep/of them could do it'), **reflexive**, performed like personal pronouns but often more emphatically, **reciprocal** (e.g., 'They did it to each/double pointing sweep/ other').

The movement indicating **pluralization** is not determined by the plural pronoun or adjective, but by the spatial relationship described in the sentence. For instance, '/Several people spoke/' would be a hand-sweep movement, but '/There are several people in that car/' would not. So, we cannot classify 'several' as a pluralization marker in itself.

I might add that Birdwhistell (1970) differentiated 'pronominal markers' (which, as has been seen, are also adjectives), 'pluralization markers', 'verboid markers', 'area markers', and 'manner markers', while nouns, adjectives, adverbs, prepositions and conjunctions do not appear in his classification. Yet, **nouns** appear under most kinesic 'illustrators', **adjectives** appear as 'identifiers' of quality and quantity, a great number of **adverbs** and adverbial constructions appear also as 'identifiers' of manner ('However' being an emphatically marked one, usually with a raised hand and typically dominated by its quality as 'identifier'), as 'time markers', as 'space markers', etc. **Prepositions** (mainly 'but') and **conjunctions** (mainly 'and') could have been clearly identified, since mainly facial and hand gestures (along with paralinguistic features) accent those linking words, quite typically coinciding with equally typical stress, pitch, volume and length features, enhancing their syntactical value and, naturally, functioning as syntactical language markers, the principal feature for a kinesic or paralinguistic act to qualify as a language marker.

Paralinguistic-kinesic stress markers must also be acknowledged in the study of speech. In the sentence 'My *wife* doesn't think *he* did it, but *I* do', there is a secondary stress on *wife* and primary stresses on *he* and *I*. Stresses are characterized by higher pitch and volume and slightly tenser articulation and longer syllabic length. By themselves those paralinguistic features are truly markers that enhance the fact that 'my wife' (not everybody, nor the two of us) thinks that way, that she doesn't think he (among other possible suspects) did it, and that I myself (unlike her) think he did. One can speak the same words in an unspirited, monotone voice, devoid of markers, and one could also (e.g., in a deep depressive state) strip them of any kinesic accompaniments; or use paralinguistic markers but not any conspicuous kinesic behaviors. Under normal circumstances, however, there are kinesic stress markers, discussed first by Birdwhistell (1970), that would coincide with phonetic stresses on the same syllables, 'wife', 'he', and 'I', as an orientation head movement toward her (if she is present) or a brow raise with 'I'. The paralinguistic-kinesic compound

appears, therefore, as the coherent construct perfectly costructured with verbal utterances as part of the Basic Triple Structure.

As for **punctuation markers**, both paralinguistic and kinesic markers truly punctuate the verbal sentence as clearly as we punctuate a written one. Again, a normal-speed or frame-by-frame analysis of a videotaped conversation reveals the precise correlation of paralinguistic and kinesic markers with written punctuation symbols, after all invented in an attempt to represent or evoke the reality of speech. Punctuation symbols represent audible features either overriding ([!] for volume, [" "] for emphasis) or momentary ([,] [:] [.] for juncture pitch registers), and they all have their corresponding kinesic markers.

The three types of punctuation elements are represented by language markers as follows: **syntactical**: [,] ('No, it's not important, but it has to be done', with momentary lower-register + head movement arrest with first comma, lower and sustained with the second comma), [:] ('The four rules are: first - - - ', falling-pitch + juncture + head-movement arrest and brow raise with semicolon), [, ,] [,] [( )] or [- -] ('One day, before we knew each other, he - - - - ', lower-volume-and-pitch juncture at commas + overriding lower pitch between them + lower-volume-and-pitch + movement arrest at interpolating symbols + overriding lower pitch between them); **quantitative**: [- - -] [. . . .] ('But I thought . . . . ', lower/higher-pitch juncture and breathing pause + lower/higher brow raise + open mouth and eyes, all indicating suspension); **qualitative**: [" "] ('We said we "would think it over", you know?', slightly-raised pitch and head, with/without marking quotation with both hands), [!] ('¡I'm so glad you came! ', overriding higher pitch and volume for loud voice; or lower volume and pitch when [!] indicates indicates intensity, as in '¡I don't want you here! '), [?] ('¿Are you coming?', raising pitch, volume and brow toward interrogation end), (see Poyatos 1981b, on punctuation).

SILENCE AND STILLNESS AS COMPONENTS OF THE BASIC TRIPLE STRUCTURE

The relevance of silence and stillness has been gradually becoming more apparent to me in view, first, of the treatment given the so-called 'unfilled pauses' in speech, and also because of the concept of 'zero sign' in semiotics, that is, the absence of a given sign. Both seem to point at a supposed lack of significant and, therefore, of signified, that can be proved to be rather fallacious. Recently (Poyatos 1982a) I have elaborated on the semiotic, semantic, and structural significance of silence and stillness as constructs which are combined in interaction mainly with the three cosystems language-paralanguage-kinesics, as systems in themselves, but also with any other communicative modality. With relation to language, silence is the nonactivity that limits segments of audible utterances travelling over the vocal/narial-auditory

channel, marking their beginning, duration and end, and linking them or putting a stop to them. With relation to movement, particularly external one, or kinesics, the counterpart of silence is stillness. But, far from lacking meaning and the ability to perform structural and communicative functions, the nonactivities of silence and stillness constitute a form of significant sign production which is central to the study of social interaction and, therefore, to a typology of somatic signs. In sum, if sound and movement have a preponderant role in our communication system, their absence must be recognized as part of that system, or rather, as a system in itself, particularly when they are consciously encoded and decoded in interaction.

As for the mechanism of interaction, where the repertoire of somatic signs reaches its highest degree of complexity, one should bear in mind, (a) that silences, as truly noncommunicative 'unfilled' pauses (but 'pause' would not be the correct term in this case), occur only before, between, and after interactive encounters between speaker and listener, that is, as noninteractive periods (unrelated to the so-called interactive turns: speaker's turn and listener's turn); and (b) that, other than true noninteractive silences, breaks are always linguistic, paralinguistic, or kinesic true pauses within or between turns, because when one of the activities is interrupted, the other two, or at least one of them, will certainly fill that gap; hence, the important semantic and structural interrelationships of language, paralanguage, and kinesics among themselves and with silence and stillness. Furthermore, unlike communicative linguistic-paralinguistic-kinesic pauses, noncommunicative silence and stillness cannot be said to be limited at both ends by predictable behaviors (i.e. signs), nor support, emphasize, or contradict the verbal and nonverbal messages, nor show any specific structure or functions, since they happen outside communicative encounters. By exclusion, then, this kind of nonactivity falls outside the former definition of sign-carrying silence and stillness and the concern of a sign typology.

As for their functions in social interaction, both silence and stillness can act as **sign propers** with different designata, not necessarily as zero signs for absent sensible sign constructs. When we feel that 'there is in her silnece/stillness all the anguish of the situation', we do not interpret it as a substitute for verbal or nonverbal expressions any more than we do the static pose of a ballet dancer, or the puzzling raised hand of our interlocutor before his shocking statement. Certain interactive pauses and motionless moments of varying lengths are nothing except what we apprehend in that absence of sound or movement, and we do not have to see them as 'replacements' for blushing, tears, or shivering, nor for language, paralanguage, or kinesics. But silence and stillness can also act as **zero signs** and signify by the absence of sound (cf. Sebeok 1977a:118 on animal zero signs) or movement, as the witting silence with which we avoid saying something, the motionless stance of expectancy or shock, or the comedian's feedback-seeking pause. Thirdly, quite a neglected aspect of silence and stillness – particularly the former, which deserves much study within

disciplines like philosophy, semiotics, psychology, and linguistics – is their capacity to act as **carriers** of the activity just perceived, enlarging it and making it more conspicuous and better defined in a sort of instant replay that fades away afterward (but, depending on its impact, even overlapping the following sensible signs). If a long pause in conversation follows a rotund 'You will not!', this negation prolongs itself more intensely in our minds than if our interlocutor continues to speak; a grotesque cry, a guffaw, a threatening gesture, or a gentle, loving word or movement, are more so if they are followed by silence and stillness respectively, as the nonactivities serve as receptacles that are filled with the reechoing sensations of the soon vanishing signs (which, through memory, may still reecho back during future silences).

Finally, a sign typology must acknowledge the further complexity of the sign web, as the signs of the nonactivities (and one is tempted to doubt the correctness of this negative term) are always qualified and, therefore, granted a measurable dimension by their **costructuration** with preceding, simultaneous or succeeding signs, the **intensity** of the cooccurring activities, (whether sensible, such as kinesics, audible respiration, perspiration, or mental, like the tense evocation of a preceding emotional outburst, or the confused thoughts, during silence or stillness, about a tragic event), and their **duration**, which is ultimately related to both costructuration and intensity.[7]

THE JOINT TRANSCRIPTION OF LANGUAGE, PARALANGUAGE, AND KINESICS

I first discussed and practiced the transcription of the Basic Triple Structure with a joint team from the Universities of Birmingham and Nancy in 1976. It was inspired to me in the late sixties by the classic in the field, **The First Five Minutes**, the pioneering work of the psychiatrists Pittinger and Danehy and the linguist-anthropologist Hockett (1960) on a five-minute segment of a psychiatric interview, where, mainly by means of arbitrary graphic symbols, they represented the linguistic and paralinguistic behaviors of the patient, adding to it a minute analysis of what goes on during the interview. To that I added the third element of the Triple Structure, kinesics, for a realistic depiction of the speech-and-movement occurrence.

The characteristics of the transcription are indicated in Figure 3, in what is actually the beginning of a blank transcription sheet showing the order in which the different features are incorporated from top to bottom, this first sheet being followed by successive ones.

1. **Orthographic transcript** that can be read quickly by turning the pages of the transcription.

2. **Paralinguistic transcription**, comprising four types of speech phenomena: (a) **primary qualities**, or overriding person-identifying features with three degrees

above and below a base line: general timbre, volume, resonance, tempo, pitch level, intonation range, syllabic length, and rhythm, plus any of the qualifiers listed as the next paralinguistic category which might also be part of the person's permanent voice set (e.g., poor articulatory control or creaky voice due to laryngealization); any of these overriding features can be indicated by a capital initial and degree (e.g., Rn3 for very nasal resonance), (b) **qualifiers**, as nonpermanent features among: respiratory control, glottis control, laryngeal-, velar-, pharyngeal-, articulatory-, labial-, and maxillary-controls, and articulatory tension; and, within the same band of the transcription, (c) **differentiators**, when they override words (otherwise included within segmental alternants below): laughter, crying, sighing, yawning, sneezing, belching, hiccoughing and extremes of whispering and loud voice; and (d) independent segmental **alternants**, such as clicks, moans, hesitation vowels, narial ingressions and egressions, etc., and including pauses.

For the paralinguistic part of the transcription the researcher may resort to what symbols may seem best among the International Phonetic Alphabet, those used by Trager (1958), Pittinger *et al* (1960), Crystal and Quirk (1964), Austin (1965), McQuown *et al* (1971), or Poyatos (1975).

3. **Phonemic transcription**, including intonational features, using mainly the IPA (International Phonetic Alphabet).

4. **Kinegraphic and parakinegraphic transcription**, completing the Basic Triple Structure, using either Birdwhistell's (1970) kinegraphs, or from a more practical point of view, the abbreviated system suggested by Kendon (1969), consisting of: eyes, brows and forehead, mouth, head, hand and arms, shoulders and trunk, and direction of gaze. Based on these two systems, one could simplify the recording of body movements and positions by differentiating, as shown in the table, three principal areas: face (eyes, brows, cheeks, nose, mouth), head-trunk-legs-feet, and shoulders-arms-hands (arms, forearms, wrists, angle, whole hand, fingers). **Parakinesic** nonsegmental features, that is, intensity, range, and velocity of movements, are recorded for specific activities but they can also appear as the permanent person-identifying features of an individual's body set, indicated, like voice set, at the beginning of the kinesic transcription.

5. **Adaptors**, that is, body-adaptor behaviors (mainly clothes and jewelry, masticatories, etc.) and object-adaptor ones (occurring either throughout the speaker's deliverance or momentarily must be indicated at this level closest to kinesics (fidgeting, tapping, chewing, etc.).

6. **Proxemic notation**, for spatial features and shifts that are intimately built in the interaction and associated with language, paralanguage, and kinesics, can follow Hall's (1974) technique or one's own according to the aims of the study, so that the roles played in the interaction by interpersonal intimate or personal distance are clearly acknowledged.

Orthographic transcript				A
				B
Primary Qualities	A. Speaker/Listener	B. Listener/Speaker	Qualifiers	A
				B
			Differentiators	
			Alternants	A
				B
Phonemic transcription				A
				B
Parakinesic Qualities	A. Speaker/Listener	B. Listener/Speaker	Face: eyes brows, forehead nose, cheeks mouth	A
				B
			Head trunk legs, feet	A
				B
			Shoulders arms, hands fingers	A
				B
Body-adaptors Object-adaptors				A
				B
Proxemic behavior				A
				B
Contextual description				A
				B

Figure 3. Triple Transcription-Description (chronemic features added to each behavior)

7. **Chronemic notation**, added to each of the other sections of the triple and proxemic notations, indicates the significant temporal features of the activities registered in all the other sections after primary qualities; that is, the duration of qualifiers like labial control and whispering, of differentiators like laughter, of alternants like independent laughter, moaning sounds and pauses; then the length of certain gestures, manners and postures (manners constituting the 'formative' and 'releasing' phases of postures) and of the various parts of the speaker's turn, of the turn itself and of the whole speaker-listener exchange.

8. **The contextual description** implies that, regardless of the aims of the research, the transcription cannot be limited to a symbolization of physical events in a rather 'desocialized' or 'deculturized' way, and so we must also indicate among other conditioning factors the cultural environment as well as the cultural background of the participants, the nature of the encounter, the personal characteristics of the participants, both physically (appearance, dress and grooming, etc.) and psychologically, their socioeconomics and socioeducational status and, of course, the type of relationship that determines the encounter. One characteristic of the proposed transcription is that both the speaker's and the listener's behaviors should be recorded simultaneously, since the interaction is at all moments a two-way affair. The listener's feedback and the speaker's simultaneous turns and interruptions, among other behaviors within the mechanism of conversation, must be recorded for a proper understanding of the encounter.

NOTES

1. In a recent study of punctuation as nonverbal communication (poyatos 1981b) I have discussed the illogical use in English of [¡] and [?] only at the end of a sentence, when the feature it represents may override the whole sentence (in which case the Spanish practice [¡ - - - - - !], and always [¿ - - - - - ?] would seem more appropriate), the first part of it (when [¡ - - - - !] could be used), or only the second one (when English ! would truly represent reality). In this article I use both [¡ - - - - - !] and [¿ - - - - - ?] as a suggestion for a more realistic representation of speech.
2. Although the terms behavior and behavioral activity are used, it must be understood that not only are controllable behavioral (a gesture, a silence, a throat clearing) and uncontrollable nonbehavioral (blushing, tear shedding) activities costructured, but those sensibly perceived and mental ones as well (e.g., my recollection of an event eliciting tear shedding or quavering voice) and my cointeractant's. The extremely intricate web of obvious and hidden relationships in interaction cannot be neglected, nor naively reduced (see the forty-one channels of intersomatic direct and synesthesial perception in Poyatos 1983:

54-56, and the definition of free and bound activities in Note 5 below).

3. I coined the term 'chronemics' (as analogous to proxemics) in the early 70s (as noted in Key 1980) as the conceptualization and handling of time as a biopsychological and cultural element lending specific characteristics to social relationships and to the many events contained within the communicative stream, from linguistic syllables and flitting gestures through meaningful glances and silences to the duration of a whole encounter.
4. Besides other efforts towards an understanding of voice production and qualities (e.g., Catford 1964, 1968 Crystal 1971, Laver 1974), psychological and clinical workers have made great strides in this area (e.g., Vetter 1969, Travis 1971, Darby 1981; see Scherer 1982 on methods in psychology), as has Laver recently (Laver and Hanson 1981 and, above all, Laver 1980). But many researchers seem to ignore those efforts and a workable classification of paralinguistic phenomena is still needed for the study of the structure of conversation, of clinical or employment interviews, the theater, advertising, 'cultural styles', etc.
5. In a recent paper dealing with the semiotic-communicative problems of intercultural interaction as well as first-language *vs.* foreign-language acquisition and use (Poyatos 1984a) I have elaborated further on the concepts 'free' and 'bound', as they must be understood at a still much deeper level, including behavioral and nonbehavioral activities (as well as the nonactivities of silence and stillness) and mental processes, all of which are subject to the interactive coding processes and costructuration outlined earlier. However, as we progressively deepen our knowledge in this area we discover, not just nonbehavioral objectual or environmental bound elements in interaction or noninteraction (e.g., paralanguage and clothes, lexical choice and the lower educational status betrayed by wall decorations hung too high) but the sensible and intelligible ways in which we can observe those interrelationships (e.g., moral and religious values bound to the wearing of clothes) and that ultimately it is hard to conceive of a free activity. At the level of the basic language-paralanguage-kinesics structure, however, and of the basic sign systems, important research applications can be developed in many areas, beginning logically with the developmental study of language, which should be of language-paralanguage-kinesics.
6. Among the established nonverbal categories (see Poyatos 1983: Chapter 4, and 1984b), the two most important ones in discourse, along with language markers, are: **identifiers**: kinesic or kinesic-paralinguistic expressions of abstract concepts ('impossible', 'absurd'), moral and physical qualities ('tough', 'cautious', 'hard', 'unfriendly') and qualities of objectual and environmental referents ('dirty', 'hard', 'soft', 'murky'); and **externalizers**: reactions to other people's past, present, anticipated or imagined reality, to what has been said, is being said or will be said, silenced, done or not done by us or someone else, to

past, present, anticipated or imagined events, to our own somatic phenomena, to animal and environmental agents, to esthetic experiences and to spiritual experiences. Externalizers do not illustrate words, as identifiers do, but are at most reactions to them. The three of them are responsible for the specific expressive style of a person (usually referred to vaguely as 'elegant', 'ladylike', 'ebullient', 'vulgar', and the like) and a whole culture (which allows us to identify it without even hearing its language).

7. Silence has also a number of instrumental or accidental interactive functions in conversation, e.g.: turn-opening, transitional, feedback-seeking, task-performing (smoking, drinking, preening), thinking, external interference, manipulation of the other person, etc.

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